SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1706**

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.* (BTL 5)

Assessment Tool Weightage: 50%

Duration : 1 Hour
QUESTIONS: 5 **Total Marks:50**

Annexure A

Technical Presentation

Code of Conduct:

I _SR Kaarthika(192412294)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SR Kaarthika

Annexure A

Technical Presentation

1. Reactive Agent-Based Traffic Control System
Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis
Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. Utility-Based Agent for E-Commerce Recommendations
Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. Learning Agent for Personalized Education System
Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

5. Multi-Agent System for Disaster Management
Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

ANSWERS:

1. Reactive Agent-Based Traffic Control System

Introduction

A Reactive Agent is an intelligent agent that responds directly to changes in its environment. In a smart traffic control system, it continuously monitors real-time traffic conditions and changes traffic signals accordingly.

Problem

Fixed-time traffic signals may cause unnecessary waiting and congestion when traffic density changes. A reactive agent can respond immediately to changing traffic conditions.

Objectives

- Monitor traffic in real time.
- Detect traffic density.
- Dynamically control traffic signals.
- Reduce congestion and waiting time.

System Architecture

Traffic Sensors/Cameras → Streaming Data → Reactive Agent → Traffic Analysis → Signal Controller → Traffic Signals

Working

1. Sensors and cameras collect vehicle-density information.
2. Streaming data is processed continuously.
3. The reactive agent checks the current traffic condition.
4. If one road has heavy traffic, the agent increases its green-signal duration.
5. When traffic decreases, the signal timing is adjusted again.

Role of PySpark

PySpark can process large amounts of streaming traffic data in a distributed manner. It helps analyze traffic information from multiple roads simultaneously.

Agent Decision

High Traffic → Increase Green Time
Medium Traffic → Normal Signal Time
Low Traffic → Reduce Green Time

Advantages

- Reduces traffic congestion.
- Provides real-time response.
- Improves road utilization.
- Reduces unnecessary waiting.

Conclusion

A Reactive Agent-based traffic control system continuously responds to real-time traffic inputs and dynamically manages signals to minimize congestion. The assessment specifically requires explaining real-time inputs, signal changes, and streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis

Introduction

A Goal-Based Agent makes decisions by considering a specific goal. In healthcare, the goal can be to identify the most suitable diagnosis and select an appropriate treatment pathway based on patient information.

Problem

Healthcare systems must analyze different patient data and select suitable diagnostic and treatment decisions.

Objectives

- Collect patient information.
- Set diagnostic goals.
- Analyze patient data.
- Select an appropriate treatment pathway.

System Architecture

Patient Data → Data Pipeline → Goal-Based Agent → Diagnosis Analysis → Decision Making → Treatment Pathway

Working

1. Patient information is collected.
2. The data is processed through a data pipeline.
3. The agent establishes a diagnostic goal.
4. It evaluates symptoms and available patient information.
5. Different possible decisions are compared.
6. The agent selects the pathway that best satisfies the defined goal.

Decision-Making Logic

The agent follows:

Patient Data → Possible Diagnoses → Goal Evaluation → Best Decision → Treatment Recommendation

Role of Data Processing

Large amounts of healthcare data can be processed using distributed data-processing techniques. This allows the system to handle information efficiently.

Advantages

- Goal-oriented decision making.
- Faster analysis of patient information.
- Supports healthcare professionals.
- Can process large datasets.

Conclusion

A Goal-Based Agent uses diagnostic goals, patient data, and decision-making logic to select suitable healthcare pathways. The assessment specifically requires goal setting, patient-data evaluation, and optimal treatment-pathway selection.

3. Utility-Based Agent for E-Commerce Recommendations

Introduction

A Utility-Based Agent selects an action by comparing the usefulness or utility value of different choices. In e-commerce, it can recommend products that maximize user satisfaction.

Problem

Online stores contain thousands of products, making it difficult for users to identify the most suitable products.

Objectives

- Understand user preferences.
- Assign utility values to products.
- Compare different product options.
- Recommend the most useful products.

System Architecture

User Data → PySpark Processing → Product Analysis → Utility Calculation → Utility Agent → Personalized Recommendations

Working

1. The system collects user searches, purchases, ratings, and preferences.
2. Large-scale user and product data is processed.
3. The agent calculates utility values for different products.
4. Products are compared based on their utility.
5. The products with higher utility are recommended.

Example

Suppose a user is interested in laptops. The agent may consider:

- Price
- Rating
- Features
- Previous user preferences

It calculates the overall utility and ranks suitable laptops.

Role of PySpark

PySpark can process large-scale customer and product data in parallel, allowing recommendations to be generated efficiently**.**

Advantages

- Personalized recommendations.
- Improved customer satisfaction.
- Efficient processing of large datasets.
- Better product selection.

Conclusion

A Utility-Based Agent improves e-commerce recommendations by assigning utility values to available options and selecting products that maximize user satisfaction.

4. Learning Agent for Personalized Education System

Introduction

A Learning Agent improves its performance by learning from feedback and previous experiences. In an educational system, it can adapt learning content according to individual student performance**.**

Problem

All students have different learning abilities and may require different levels of content and practice**.**

Objectives

- Monitor student performance.
- Learn from feedback.
- Adapt content difficulty.
- Provide personalized learning.

System Architecture

Student Activity → PySpark → Performance Analysis → Learning Agent → Content Adaptation → Student Feedback → Learning

Working

1. Student activities and performance data are collected.
2. PySpark processes large amounts of learning data.
3. The agent analyzes student performance.
4. Feedback is used to evaluate the effectiveness of the current content.
5. The agent changes the difficulty or type of content.
6. The learning process continues and the agent improves over time.

Learning Process

Student Performance → Feedback → Agent Learning → Content Adjustment → Improved Learning

Role of PySpark

PySpark helps process distributed student data from many learners efficiently.

Advantages

- Personalized learning.
- Adaptive difficulty.
- Continuous improvement.
- Real-time feedback.
- Supports large numbers of students.

Conclusion

A Learning Agent provides personalized education by learning from student feedback and adapting content difficulty. The assessment specifically requires feedback-based improvement, content adaptation, and integration with PySpark.

5. Multi-Agent System for Disaster Management

Introduction

A Multi-Agent System consists of multiple intelligent agents that work together to solve a complex problem. In disaster management, different agents can monitor, analyze, communicate, and coordinate emergency responses.

Problem

Disaster situations require fast detection, information sharing, and coordinated decision-making.

Objectives

- Detect disasters quickly.
- Collect information from multiple sources.
- Share information between agents.
- Support real-time decision-making.

System Architecture

Sensors → Sensor Agents → Decision Agent → Communication Agent → Emergency Response

Types of Agents

1.SensorAgent
Collects information such as environmental conditions and disaster-related sensor readings.

2.DecisionAgent
Analyzes the collected information and determines the appropriate response.

3.CommunicationAgent
Shares warnings and decisions with authorities, emergency teams, and other agents.

Working

1. Sensor agents continuously collect data.
2. The collected information is shared with decision agents.
3. Decision agents analyze the situation.
4. Communication agents distribute alerts.
5. Agents coordinate their actions to support disaster response.

Coordination

Data Collection → Data Sharing → Decision Making → Alert → Response

Role of Distributed Processing

Large amounts of disaster-related data can be processed using distributed systems such as PySpark, allowing multiple data sources to be analyzed efficiently.

Advantages

- Fast disaster detection.
- Real-time decision-making.
- Effective communication.
- Cooperation between multiple agents.
- Better emergency response.

Conclusion

A Multi-Agent System provides an effective disaster-management solution by allowing sensor, decision, and communication agents to collaborate. The assessment specifically emphasizes coordination, data sharing, and real-time decision-making.